IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SELECTED TOPICS IN POWER SYSTEMS

Wednesday, 6 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Question 1 on Flexible AC Transmission System (FACTS)

a) Compare the *ideal* phase angle compensation achieved using a Thyristor Controlled Phase Angle Regulator (TCPAR) versus the phase angle compensation provide by a Quadrature Booster (QB) in terms of the following aspects. Consider each compensator to be installed in a transmission line with equal voltage magnitudes at the two ends and both positive ($+\sigma$) and negative ($-\sigma$) phase shift scenarios. Neglect line resistance and shunt susceptance.

i) Phasor diagram - Draw and label a phasor diagram for TCPAR and QB.

[2 marks]

ii) Power-angle characteristics - Draw and label how the active power transfer through a transmission line varies with the phase angle for both TCPAR and QB, in comparison to the uncompensated (base case) scenario.

[2 marks]

iii) Injected voltage - Explain the nature of the voltage injected by each device for phase angle compensation.

[2 marks]

iv) Key components - Mention the key components that differentiates the operation of a TCPAR and QB.

[2 marks]

b) A TCSC (Thyristor Controlled Series Capacitor) is installed on a transmission line with a reactance of $X = 0.3$ p.u. Assume the line has negligible resistance and shunt susceptance, and that the sending- and receiving-end voltages are both maintained at 1.0 p.u. Answer the following:

i) What series compensation level (k in %) must the TCSC provide to double the line's maximum active power transfer capacity compared to the uncompensated case?

[2 marks]

ii) Determine the critical clearing angle (in degrees) if the TCSC provides zero compensation ($k = 0$) and the pre-fault active power flow is 2.0 p.u. Assume identical line reactance pre- and post-fault and active power transfer is zero during the fault.

[4 marks]

iii) Recalculate the critical clearing angle (in degrees) if the series compensation level of the TCSC is $k = 0.3$ under the same assumptions as in part (ii).

[3 marks]

iv) If the critical clearing angle were to be the same as that obtained in part (ii), what is the maximum allowable pre-fault active power that can flow through the line when $k = 0.3$?

[3 marks]

[20 marks]

Question 2 on System Operation with Low Inertia

a) To maintain frequency stability in the future low-inertia power system, it is important to meet the system frequency requirements after large disturbances. The system dynamics are complicated but can be approximately described by the following swing equation:

$$2H\dot{f}(t) = R(t) - \Delta P_L$$

i) Please list FIVE potential stability and security challenges caused by inverter-based resources

[3]

ii) Analytically derive the frequency nadir (at time instant t_n) constraint in terms of H (system inertia in [s]), R (frequency response linearly delivered by T_d) and Δf_{max} from the swing equation above and assume $t_n < T_d$.

[3]

iii) The constraint that guarantees the system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(H - \mathrm{DC}\times\frac{\mathrm{T_{DC}}}{4\times\Delta f_{\max}}\right)\times R \geq \frac{(P_{\mathrm{L}} - \mathrm{DC})^2 \times \mathrm{T_d}}{4\times\Delta f_{\max}}$$

Nomenclature

H: System level of inertia (MWs/Hz)
R: Amount of frequency response scheduled (MW)
DC: Amount of Dynamic Containment (also called Enhance Frequency Response (EFR)) scheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
$\Delta f_{\max}$: Maximum admissible frequency deviation to avoid load shedding
$\mathrm{T_d}$: Delivery time of frequency response (s)
$\mathrm{T_{DC}}$: Delivery time of DC (s)

Other assumptions:

- The largest generation unit is 1.8 GW
- Numerical values: H = 3 GWs/Hz, $\Delta f_{\max}$ = 0.8 Hz, $\mathrm{T_d}$ = 11.2 s, $\mathrm{T_{DC}}$ = 0.64 s

The system operator has procured the **optimal** amount of DC and R to maintain the nadir constraint. In this scenario, it happens that $\boldsymbol{DC = 0.55\ \mathrm{GW}}$. Calculate R that have been procured and the relationship of their marginal prices.

[4]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) Explain what control mechanisms a variable-speed wind turbine can use to create a headroom (ie be able to ramp up its active power when required) to provide frequency-support services.

[4]

ii) A wind turbine with a power rating of 8 MW operates above rated wind speed with a deloading of 1 MW (ie it generates 7 MW). Calculate the highest synthetic inertia constant, H, the wind turbine could emulate under this operating point if the worst RoCoF expected is of -2 Hz/s and the system's nominal frequency is 50 Hz.

[3]

iii) Draw the approximate graph of the kinetic energy stored in a wind turbine versus the wind speed. Explain the shape of the curve.

[3]

[20 marks]

Question 3 on Power System State Estimation

a)

i) Why is state estimation necessary in power systems? [4]

ii) What is weighted least square method of state estimation? [3]

iii) What are its advantages and disadvantages of this method? [3]

b) In the context of power distribution system

i) Why is state estimation not a routine task? [2]

ii) How are the measurements classified for state estimation? [3]

iii) How are these measurements represented in state estimation objective function? [3]

iv) Why is it not straight forward to include data from smart meters for state estimation? [2]

[20 marks]

Question 4 on Integrated Energy Systems

a) Part I

Describe the challenges for the power system in deep electrification of energy demand and how this can be addressed?

[3]

How can hydrogen facilitate the decarbonsination of heat (provide at least two applications)?

[2]

b) Part II

Considering the electrification of the heat sector, heat appliances through thermal energy storage (TES) could provide flexibility to the electricity system and support consumer energy bills. Consider a system with a three-hour snapshot including the grid electricity price ∂_t^{Grid}, electricity demands of P_t^D and the heat demand of H_t^D, as shown below in the figure.

- The employed CHP produces the same heat and electricity output, where the generation cost is ∂^{CHP} =8p/kW$_{el}$.
- The thermal energy storage (TES):
  - Maximum energy capacity of 10kWh.
  - TES is empty in t=0 and the energy at t=3 is 8kWh. Furthermore, it is assumed that at t=1, the TES will be charged for 6kWh.
  - Charging rate of 6kW and discharging rate of 4kW (i.e., in binary, meaning it is either 0 or 6kW (for charging) and 4 kW (for discharging)).
  - The heat loss rate is ignored, and the TES cannot be charged and discharged at the same time.

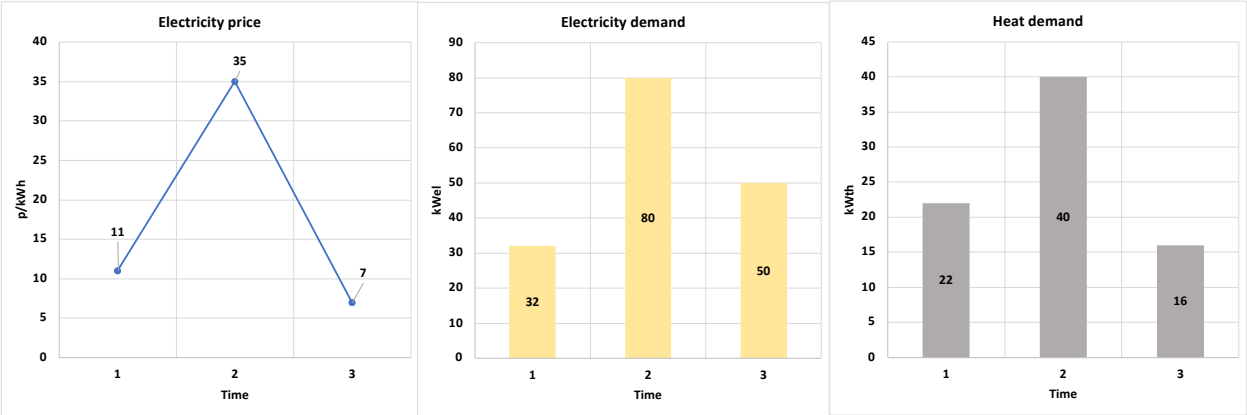


Figure 4.1 grid electricity price ∂_t^{Grid}, electricity demands of P_t^D and the heat demand of H_t^D

The objective is to minimise the consumers energy bills:

$$Obj = \min \sum_{t=1}^{3} \left(\partial_t^{\text{Grid}} \cdot P_t^{\text{Grid}} + \partial^{\text{CHP}} \cdot P_t^{\text{CHP}} \right)$$

The following equations on electricity and heat balances are provided:

$$P_t^D = P_t^{\text{CHP}} + P_t^{\text{grid}}, t = 1,2,3$$

$$H_t^D = H_t^{\text{CHP}} + H_t^{\text{TES,Dis}} - H_t^{\text{TES,Ch}}, t = 1,2,3$$

$E_t^{\mathrm{TES}} = E_{t-1}^{\mathrm{TES}} + H_t^{\mathrm{TES,Ch}} - H_t^{\mathrm{TES,Dis}},\ t = 2,3$

i) Assuming that TES is not available, what is the total consumer energy bill during the provided time period?

[3]

ii) Now, the TES can provide flexibility to the energy system, calculate the minimum consumer energy bill. Provide the full calculations.

[8]

iii) Calculate the total savings in presence of TES on the consumer bill in %:

[1]

iv) Assuming the electricity price grid at t=2 is not known: $\partial_t^{\mathrm{Grid}}$= (11,X,7). What should be the price of electricity at t=2, so that charging/discharging of the TES does not impact the consumer bill.

[3]

[20 marks]